

A Boundary Reduction for Type (ANA) Maximally Monotone Linear Relations

Automatic math-discovery smoke test

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Source Problem

The source paper is Bauschke–Borwein–Wang–Yao, *Construction of pathological maximally monotone operators on non-reflexive Banach spaces*, arXiv:1108.1463. Near the end of the paper the authors list four open questions:

$$\boxed{\text{type (D)}} \longleftrightarrow \boxed{\text{type (FP)}} \longleftrightarrow \boxed{\text{type (AN)}} \longleftrightarrow \boxed{\text{type (LD)}} \longrightarrow \boxed{\text{uniqueness}}$$

The following four questions are left open.

- (i) Is every maximally monotone operator necessarily of type (FPV)?
- (ii) Is every maximally monotone operator necessarily of type (ANA)?
- (iii) Is every maximally monotone linear relation necessarily of type (ANA)?
- (iv) Is every maximally monotone operator of type (BR) necessarily of type (ANA)?

The first of these is especially important, being closely related to the sum theorem in general Banach space (see [35, 11, 9, 40]).

This packet concerns question (iii):

Is every maximally monotone linear relation necessarily of type (ANA)?

Here type (ANA) means “almost negative alignment”. In the notation of Simons [3, definition labeled ANAdef], a maximally monotone subset $A \subset E \times E^*$ is of type (ANA) if, whenever $(w, w^*) \notin A$, there are graph points $(s, s^*) \in A$ with $s \neq w$, $s^* \neq w^*$, and

$$\frac{\langle s - w, s^* - w^* \rangle}{\|s - w\| \|s^* - w^*\|}$$

as close to -1 as desired.

Result Type

This is a **partial result**. It proves that any possible failure of type (ANA) for a maximally monotone linear relation must occur at a boundary point of its domain. In particular, it proves the source question for all maximally monotone linear relations with closed domain.

Idea of the Proof

Let $A : X \rightrightarrows X^*$ be a maximally monotone linear relation, put $D = \text{dom } A$, and let $M = \overline{D}$. Maximality gives

$$A(0) = M^\perp.$$

Thus the fibers of A , after restriction to M , are single-valued: there is a linear monotone operator

$$T : D \subset M \rightarrow M^*, \quad Tx = x^*|_M \quad (x^* \in Ax),$$

and each fiber Ax consists exactly of all Hahn–Banach extensions of Tx from M to X .

The ANA construction then splits into two robust cases. If $w \notin M$, the positive quotient distance from w to M produces a norming annihilator $n^* \in M^\perp$; adding a large multiple of n^* in the fiber forces the angle to tend to -1 . If $w \in D$, a small-scaling argument on the normed space D gives negative alignment for the induced single-valued operator, and Hahn–Banach lifts it back to X^* without changing the numerator or the relevant norm.

The only place where both mechanisms disappear is the genuine boundary case $w \in \overline{\text{dom } A} \setminus \text{dom } A$.

Main Reduction

Theorem 1 (Boundary reduction). *Let X be a real Banach space and let $A : X \rightrightarrows X^*$ be a maximally monotone linear relation. Put $D = \text{dom } A$ and $M = \overline{D}$. Then A has the type-(ANA) conclusion at every $(w, w^*) \in X \times X^* \setminus \text{gra } A$ such that either*

$$w \notin M \quad \text{or} \quad w \in D.$$

Consequently, any counterexample to the assertion that every maximally monotone linear relation is of type (ANA) must fail at a point

$$w \in \overline{\text{dom } A} \setminus \text{dom } A.$$

Proof. Since A is a linear relation, monotonicity is equivalent to

$$\langle x, x^* \rangle \geq 0 \quad ((x, x^*) \in \text{gra } A).$$

We first identify the multivalued part. If $u^* \in A(0)$, then for each $(x, x^*) \in \text{gra } A$ and every $t \in \mathbb{R}$, $(x, x^* + tu^*) \in \text{gra } A$. Hence

$$0 \leq \langle x, x^* + tu^* \rangle \quad (t \in \mathbb{R}),$$

which forces $\langle x, u^* \rangle = 0$ for all $x \in D$. Thus $A(0) \subset D^\perp = M^\perp$. Conversely, if $u^* \in M^\perp$, then for every $(x, x^*) \in \text{gra } A$,

$$\langle 0 - x, u^* - x^* \rangle = -\langle x, u^* \rangle + \langle x, x^* \rangle = \langle x, x^* \rangle \geq 0.$$

Thus $(0, u^*)$ is monotonically related to $\text{gra } A$, and maximality gives $u^* \in A(0)$. Therefore

$$A(0) = M^\perp. \tag{1}$$

Define $T : D \rightarrow M^*$ by

$$Tx = x^*|_M \quad \text{for any } x^* \in Ax.$$

This is well-defined by (1). It is linear and monotone. Moreover, for $x \in D$, the fiber Ax consists exactly of all extensions of Tx from M to X : one inclusion is the definition of T , and the other follows because two extensions differ by an element of $M^\perp = A(0)$.

The induced operator T is maximally monotone as a relation in $M \times M^*$. Indeed, suppose $(y, y^*) \in M \times M^*$ is monotonically related to $\text{gra } T$. Choose an extension $x^* \in X^*$ of y^* . Then (y, x^*) is monotonically related to $\text{gra } A$, so maximality of A gives $(y, x^*) \in \text{gra } A$. Hence $y \in D$ and $y^* = Ty$.

Fix $(w, w^*) \notin \text{gra } A$ and $0 < \varepsilon < 1$.

Case 1: $w \notin M$. Let $\delta = \text{dist}(w, M) > 0$. By Hahn–Banach in the quotient X/M , choose $n^* \in M^\perp$ with $\|n^*\| = 1$ and $\langle w, n^* \rangle = \delta$. Since D is dense in M , choose $s \in D$ such that

$$\|s - w\| < \frac{\delta}{1 - \varepsilon/2}.$$

Then $\langle s - w, n^* \rangle = -\delta$. Pick any $a^* \in As$. For $\lambda > 0$, put $s_\lambda^* = a^* + \lambda n^*$. By (1), $s_\lambda^* \in As$. Consequently

$$\frac{\langle s - w, s_\lambda^* - w^* \rangle}{\|s - w\| \|s_\lambda^* - w^*\|} = \frac{\langle s - w, a^* - w^* \rangle - \lambda \delta}{\|s - w\| \|a^* - w^* + \lambda n^*\|} \longrightarrow -\frac{\delta}{\|s - w\|} < -1 + \frac{\varepsilon}{2}.$$

For large λ , this ratio is $< -1 + \varepsilon$.

Case 2: $w \in D$. Let $z^* = Tw - w^*|_M \in M^*$. If $z^* = 0$, then w^* is an extension of Tw , hence $w^* \in Aw$, contrary to $(w, w^*) \notin \text{gra } A$. Thus $z^* \neq 0$.

Because D is dense in M , the norm of z^* is computed on vectors from D . Choose $h \in D$ with $\|h\| = 1$ and

$$\langle h, z^* \rangle < -(1 - \varepsilon/2) \|z^*\|.$$

For $t > 0$, set $s_t = w + th \in D$. The corresponding functional on M is

$$Ts_t - w^*|_M = z^* + tTh.$$

Choose $u_t^* \in X^*$ extending $z^* + tTh$ from M to X with

$$\|u_t^*\|_{X^*} = \|z^* + tTh\|_{M^*},$$

and put $s_t^* = w^* + u_t^*$. Then s_t^* extends Ts_t , so $s_t^* \in As_t$. Also $s_t \neq w$ and $s_t^* \neq w^*$ for small $t > 0$. Finally,

$$\frac{\langle s_t - w, s_t^* - w^* \rangle_{X, X^*}}{\|s_t - w\| \|s_t^* - w^*\|_{X^*}} = \frac{\langle th, z^* + tTh \rangle_{M, M^*}}{t \|h\| \|z^* + tTh\|_{M^*}} \longrightarrow \frac{\langle h, z^* \rangle}{\|z^*\|} < -1 + \frac{\varepsilon}{2}.$$

For sufficiently small $t > 0$, the ratio is $< -1 + \varepsilon$.

These two cases prove the claimed type-(ANA) conclusion away from the boundary $M \setminus D$. The final statement follows immediately. $\square$

Corollary 1 (Closed-domain subcase). *Let $A : X \rightrightarrows X^*$ be a maximally monotone linear relation. If $\text{dom } A$ is closed, then A is of type (ANA).*

Proof. If $D = \text{dom } A$ is closed, then $M = \overline{D} = D$. Every point $(w, w^*) \notin \text{gra } A$ has either $w \notin M$ or $w \in D$, so the theorem applies. $\square$

Relation To Known Literature

Simons proves that every closed monotone quasidense set is of type (ANA) [3, theorem labeled **ANAtm**]. For linear relations, quasidensity is equivalent to monotonicity of the adjoint in the same paper [3, theorem labeled **LINMONtm**]. The reduction above is different: it does not assume quasidensity, and the closed-domain corollary includes non-quasidense full-domain examples such as the tail/Gossez-type operators.

The later survey [4] records that Bauschke–Simons proved arbitrary-domain maximally monotone linear operators are strongly maximal, and also that convex-graph maximally monotone multifunctions are strongly maximal. This is related but does not by itself give the type-(ANA) conclusion used here.

Novelty And Search Notes

Local run indexes were searched for arXiv:1108.1463, “type (ANA)”, “almost negative alignment”, “maximally monotone linear relation”, “closed domain”, and nonclosed-domain variants. No prior packet or attempt for this source question was found beyond the earlier version of this packet. Local source search found Simons’s quasidensity paper arXiv:1407.1100 and the 2017 survey arXiv:1711.06165. A web search for exact type-(ANA) linear-relation and boundary-domain phrases did not return an exact match.

Limitations

This packet does not settle the source question. It reduces the possible failure to the case

$$w \in \overline{\operatorname{dom} A} \setminus \operatorname{dom} A.$$

Equivalently, after replacing X by $M = \overline{\operatorname{dom} A}$, the remaining problem is the dense, non-full-domain case for a single-valued maximally monotone linear operator $T : D \subset M \rightarrow M^*$.

References

References

- [1] H. H. Bauschke, J. M. Borwein, X. Wang, and L. Yao, *Construction of pathological maximally monotone operators on non-reflexive Banach spaces*, arXiv:1108.1463, Set-Valued Var. Anal. 20 (2012), 387–415.
- [2] H. H. Bauschke and S. Simons, *Stronger maximal monotonicity properties of linear operators*, Bull. Austral. Math. Soc. 60 (1999), 163–174.
- [3] S. Simons, *“Densities” and maximal monotonicity I*, arXiv:1407.1100, J. Convex Anal. 23 (2016), 1017–1050.
- [4] S. Simons, *Quasidensity: a survey and some examples*, arXiv:1711.06165.

Human-Review Recommendation

Review as a likely valid strengthened partial result for question (iii) of arXiv:1108.1463. A full solution should focus on the dense-domain boundary case described above.